

House Price Prediction – Project Summary

Objective

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property features such as area, number of bedrooms, bathrooms, stories, parking availability, and other amenities. The project also aimed to identify the factors that have the greatest influence on house prices.

Dataset

The project used the **Housing Prices Dataset** from Kaggle, which contains **545 house records** with information about property characteristics and their corresponding prices. The dataset included both numerical and categorical features.

Data Preprocessing

The dataset was first explored to understand its structure and quality. It was found that there were **no missing values** and **no duplicate records**. Categorical features such as **main road**, **guestroom**, **basement**, **air conditioning**, **prefarea**, and **furnishing status** were converted into numerical values using **One-Hot Encoding** to make the data suitable for machine learning algorithms.

Model Development

The dataset was divided into **80% training data** and **20% testing data**. Two regression models were trained and evaluated:

- Linear Regression
- Random Forest Regressor

The performance of both models was measured using **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **R² Score**.

Results

The Linear Regression model outperformed the Random Forest Regressor by achieving a lower MAE, lower RMSE, and a higher R² Score. Therefore, Linear Regression was selected as the best model for predicting house prices in this project.

Metric	Linear Regression	Random Forest Regressor
Mean Absolute Error (MAE)	970,043	1,021,546
Root Mean Squared Error (RMSE)	1,324,507	1,400,566
R ² Score	0.653	0.612

The R² Score indicates that the model explains approximately **65% of the variation** in house prices, providing reasonably accurate predictions for this dataset.

Key Findings

The analysis revealed that **area, bathrooms, stories, parking, and air conditioning** are among the most important factors influencing house prices. Larger houses with better facilities generally have higher market values. The visualizations also showed clear relationships between these features and property prices.

Conclusion

This project demonstrates that machine learning can effectively estimate house prices using property features. The Linear Regression model produced the best performance and can assist buyers, sellers, and real estate businesses in making informed pricing decisions. Future improvements could include using larger datasets and more advanced machine learning techniques to achieve even higher prediction accuracy.